

Tutorial 2
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1.1.A college wants to store the details of a student, including the student's name, roll number, and department. The system should display the student's information.

Task:

- Create a class Student.
- Create an object.
- Display the student's details.

```
class Student {  
    String name;  
    int rollNumber;  
    String department;  
    Student(String name, int rollNumber, String department) {  
        this.name = name;  
        this.rollNumber = rollNumber;  
        this.department = department;  
    }  
    void display() {  
        System.out.println("Name: " + name);  
        System.out.println("Roll Number: " + rollNumber);  
        System.out.println("Department: " + department);  
    }  
}  
  
public class StudentDemo {  
    public static void main(String[] args) {  
        Student s = new Student("Manimala", 101,
```

```
"Computer Science Engineering");
```

```
s.display();
```

```
}
```

```
}
```

2.A library wants to keep track of books. Each book has a title, author, and price. When a new book is added, its details should be displayed.

Task:

- Create a class Book.
- Store the title, author, and price.
- Display the book details.

```
class Book {
```

```
    String title;
```

```
    String author;
```

```
    double price;
```

```
    Book(String title, String author, double price) {
```

```
        this.title = title;
```

```
        this.author = author;
```

```
        this.price = price;
```

```
    }
```

```
    void display() {
```

```
        System.out.println("Title: " + title);
```

```
        System.out.println("Author: " + author);
```

```
        System.out.println("Price: Rs. " + price);
```

```
    }
```

```
}
```

```
public class BookDemo {  
    public static void main(String[] args) {  
        Book b = new Book("Java Basics", "James", 450.00);  
  
        b.display();  
    }  
}
```

3.A college is developing a Student Registration System to store basic information about newly admitted students. For every student, the system should store the student's name and roll number. The college also wants a feature to display the student's details whenever required.

Task:

- Create a class Student.
- Store the student's name and roll number.
- Create a method display() to print the student's details.

```
class Student {  
    String name;  
    int rollNumber;  
  
    Student(String name, int rollNumber) {  
        this.name = name;  
        this.rollNumber = rollNumber;  
    }  
  
    void display() {  
        System.out.println("Name: " + name);  
        System.out.println("Roll Number: " + rollNumber);  
    }  
}
```

```

    }
}

public class StudentRegistrationDemo {
    public static void main(String[] args) {
        Student s = new Student("Manimala", 101);

        s.display();
    }
}

```

4.4. An interior designer wants to estimate the floor area of a rectangular room before placing furniture. The application should store the length and breadth of the room and calculate its area.

Task:

- Create a class Rectangle.
- Store the length and breadth.
- Create a method to calculate and display the area of the rectangle.

```

class Rectangle {
    double length;
    double breadth;

    Rectangle(double length, double breadth) {
        this.length = length;
        this.breadth = breadth;
    }

    void displayArea() {

```

```
        double area = length * breadth;

        System.out.println("Area: " + area);
    }
}
```

```
public class RectangleDemo {

    public static void main(String[] args) {

        Rectangle r = new Rectangle(10, 5);

        r.displayArea();
    }
}
```

5.5. A school wants to maintain records of multiple students. The system should be able to store and display the details of different students individually.

Task:

- Create a class Student.
- Create two student objects with different names and ages.
- Display the details of both students.

```
class Student {

    String name;

    int age;

    Student(String name, int age) {

        this.name = name;

        this.age = age;
    }
}
```

```

    }

    void display() {
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
    }
}

public class StudentDetailsDemo {
    public static void main(String[] args) {
        Student s1 = new Student("Manimala", 19);
        Student s2 = new Student("Anu", 20);

        s1.display();
        s2.display();
    }
}

```

6. A company is building an Employee Management System to maintain employee records. Each employee has their own information, and the system should display the details of multiple employees.

Task:

- Create a class Employee.
- Create two employee objects with different information.
- Display the details of both employees

```

class Employee {
    String name;
    int id;
}

```

```
double salary;

Employee(String name, int id, double salary) {
    this.name = name;
    this.id = id;
    this.salary = salary;
}

void display() {
    System.out.println("Name: " + name);
    System.out.println("ID: " + id);
    System.out.println("Salary: Rs. " + salary);
}
}

public class EmployeeDemo {
    public static void main(String[] args) {
        Employee e1 = new Employee("Manimala", 101, 40000);
        Employee e2 = new Employee("Anu", 102, 45000);

        e1.display();
        e2.display();
    }
}
```

7. After the semester examinations, a college wants to automatically calculate each student's total marks and average marks based on marks obtained in three subjects.

Task:

- Create a class Marks.
- Store the marks of three subjects.
- Calculate and display the total marks and average marks.

```
class Marks {  
    int mark1;  
    int mark2;  
    int mark3;  
  
    Marks(int mark1, int mark2, int mark3) {  
        this.mark1 = mark1;  
        this.mark2 = mark2;  
        this.mark3 = mark3;  
    }  
  
    void display() {  
        int total = mark1 + mark2 + mark3;  
        double average = total / 3.0;  
  
        System.out.println("Total Marks: " + total);  
        System.out.println("Average Marks: " + average);  
    }  
}  
  
public class MarksDemo {  
    public static void main(String[] args) {  
        Marks m = new Marks(85, 90, 80);  
    }  
}
```

```
m.display();  
}  
}
```

8. A weather monitoring station records the temperature in Celsius, but an international weather report requires the temperature in Fahrenheit. The system should perform the conversion automatically.

Task:

- Create a class Temperature.
- Store the temperature in Celsius.
- Convert it to Fahrenheit using the formula:

$$Fahrenheit = (Celsius) + 32$$

- Display both the Celsius and Fahrenheit temperatures.

```
class Temperature {  
    double celsius;  
  
    Temperature(double celsius) {  
        this.celsius = celsius;  
    }  
  
    void display() {  
        double fahrenheit = celsius + 32;  
  
        System.out.println("Celsius: " + celsius);  
        System.out.println("Fahrenheit: " + fahrenheit);  
    }  
}
```

```
public class TemperatureDemo {  
    public static void main(String[] args) {  
        Temperature t = new Temperature(25);  
  
        t.display();  
    }  
}
```